MIL-A-38202C(USAF)
7 March 1972
SUPERSEDING
MIL-A-38202B(USAF)
13 August 1971

MILITARY SPECIFICATION

AIRCRAFT ARRESTING SYSTEM BAK-12A/E32A; PORTABLE,
ROTARY FRICTION

1. SCOPE

* 1.1 This specification covers the following emergency types of portable, rotary friction, aircraft arresting systems:

BAK-12A/E32A - Single absorber, 1,200 foot runout

BAK-12A/E32A - Dual absorber, 1,200 foot runout.

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification to the extent specified herein. The applicable issues of the governmental documents shall be the issues in effect on date of invitation for bids or request for proposal. The applicable issues of the nongovernmental documents shall be those which are available in the files of the procuring activity and made a part of the invitation for bids or request for proposal.

SPECIFICATIONS

Federal

TT-V-121	Varnish, Spar, Water-Resisting
UU-T-81	Tags, Shipping and Stock
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-40	Packaging and Packing of Hand Tools
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-P-514	Plates, Identification, Instruction and Marking, Blank

FSC 1710

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MIL-C-3774	Crates, Wood; Open, 12,000- and 16,000-Pound Capacity
MIL-G-8402	Gages, Pressure, Dial Indicating, General Specification for
MIL-E-10062	Engines; Preparation for Shipment and Storage of
MIL-E-11275	Engine, Gasoline, Industrial Type, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-B-26195	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-W-38461	Webbing, Nylon, Aircraft Arresting

STANDARDS

Military

MIL-STD-12	Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents
MIL-STD-17	Mechanical Symbols
MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

DRAWINGS

U. S. Air Force

62B35556	Wire Rope Assembly, Single Leg, Pendant, Aircraft Arresting Barrier Absorber
64C33478	Loading Arbor
64C33518	Loading Tube
66D1751	Connector, 8-1/2-inch Tape, BAK-12 Arresting Barrier, Assembly of
67F2011	Aircraft Arresting System BAK-12/E32A, Permanent Surface Type, Installation of, Typical
67F2012	Aircraft Arresting System, BAK-12/E32A, Pit Type Installation of, Typical

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*	68F211	Aircraft Arresting System Dual BAK-12/E32A - Pit Type, Installation of, Typical
*	71236	Aircraft Arresting System Dual BAK-12/E32A - Permanent Surface Type, Installation of, Typical

Naval Air Engineering Center (Code 80020)

515053	Pendant Assembly, 1-1/4 Dia. Nonrotating Wire Rope

E. W. Bliss Company (Code 80336)

152-A-1092	Utility Pump Assembly
152-A-1093	Utility Hose Assembly
52-B-624-2	Bearing Removal Tool
52-B-1242	Reinstallation Kit
52-B-1244	Stake Driving Tool
52-C-963-1	Modification - Engine Base
52-C-1246-1	Accumulator Charging Kit
52-C-1247-1	Brake Bleed Kit
52-C-1248-1	Synchronizing Kit
52-D-109-1	Tape Brush Assembly
52-D-380-1	Deflector Sheave Assembly
52-D-642-1	Tape Brush and Mounting Assembly
52-W-1766	General Arrangement BAK-12 Arresting System (Soil Surface)
52-W-2080-1	Deck Sheave Assembly
52-W-2081-1	Tape Connector Assembly
* 52-W-2252-1	Deck Sheave, Left Hand
52-W-2288-16	Synchronizing Chain Roller
52-W-2291-2	Arresting Engine Assembly
52-W-2295-1	Fairlead Unit Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

* 3.2 Components. The aircraft arresting system shall consist of the following major components:

Energy absorber	3.8
Purchase element tapes	3.9
Purchase element connectors	3.10

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Hook cable	3.11
Fairlead unit assemblies	3.12
Runway edge sheaves	3.13
Deflector sheave assemblies	3.14
Fairlead tubes and couplings	3.15
Tape brush assemblies	3.16
Tape connector adapter assembly (dual BAK-12 only)	3.17
Special service tools and kits	3.18.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

* 3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality for the purpose intended.

3.4.1 Metals. Corrosion-resistant metals shall be used in the construction of the aircraft arresting system wherever protective finishing is not practical. Nonexposed lubricated metal parts and surfaces need not be given a protective coating and need not be corrosion resistant. Protective coatings shall be as specified herein.

3.4.2 Dissimilar metals. Insofar as practicable, the use of dissimilar metals in contact with each other shall be avoided. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces will be permitted. The use of dissimilar metals separated by insulating material will be permitted. Dissimilar metals are defined in MIL-STD-889.

3.5 Design. The aircraft arresting system shall be a velocity-sensitive, rotary friction type based on the split energy absorber concept. The energy absorber units shall be sited on opposite sides of the runway with no requirement for cross-runway interconnection other than the hook cable. The aircraft arresting system shall be designed and equipped to provide a normal system runout of 1,200 feet. It shall also be capable of being programmed for a 950-foot system runout. The aircraft arresting system shall be designed for soil surface, concrete pad, or permanent-pit-type installation. The components for each type of installation shall be provided as specified on the installation drawing (see 6.2). The procuring activity shall specify the following for each site where the aircraft arresting system is to be used (see 6.2):

a. Type of installation

b. Length of hook cable

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c. Span between runway edge sheaves (see 6.3)

d. Split distance between the runway edge sheaves and energy absorber (see 6.3)

e. When only a 950-foot system runout is required, and if a 1-inch-diameter hook cable is allowed for this site

* f. When dual BAK-12 is to be used

The requirements of the installation and component drawings and all details thereto (see 6.2) apply as requirements of this specification with the exceptions and additions specified herein.

3.5.1 Soil surface installation. The design of the aircraft arresting system shall permit installation on a graded soil surface using aluminum stakes and deadman anchors. Arresting systems for soil surface installations shall be in accordance with Drawing 52-W-1766 and shall consist of at least the following major components, arranged essentially as shown thereon. Additional components, including hardware, required to complete the aircraft arresting system for this particular type installation shall be as shown on Drawing 52-W-1766.

Component	See Paragraph	Number Required
Energy absorber	3.8	2
Purchase element tapes	3.9	2
Purchase element connectors	3.10	2
Hook cable	3.11	1
Fairlead unit assemblies	3.12	2
Special service tools and kits	3.17	1 set.

3.5.1.1 Reinstallation kit. When specified (see 6.2), a reinstallation kit shall be provided for each soil surface installation. The kit shall contain the anchoring hardware for the absorber and Fairlead unit assembly. The kit shall permit quick removal and reinstallation of the arresting system by avoiding the need to dig up anchoring hardware. Necessary hardware is shown on Drawing 52-B-1242.

3.5.1.2 Special expeditionary installation kits. For aircraft arresting systems intended for expeditionary type soil surface installations, it is anticipated that spare parts and small tools will not be readily available to the installing and operating crews as they would be at established airbases. When specified (see 6.2), a kit containing particular spare parts and tools

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shall be provided for those installations. The kit shall include at least a hand portable compressed nitrogen bottle complete with regulator, pressure gage, and fittings compatible with the BAK-12 accumulator. The compressed gas is needed to pressurize the static pressure system which functions to maintain a light pressure (eg, 175 ±10 psi) on the brakes and thereby holds the pretension load in the hook cable when the aircraft arresting system is in the battery position. Minimum size of bottle to be provided shall be 0.1 cubic foot for 2,000 psi or 0.25 cubic foot for 500 psi. The kit also shall contain extra spare parts, special or extra tools, and such other items as specified (see 6.2). (Drawing 52-C-1246 may be used for information purposes only.)

* 3.5.2 Installation, permanent, surface type. The design of the aircraft arresting system shall be adaptable to installation on a simple concrete slab poured at the time of installation. Arresting systems for permanent surface type installations shall be in accordance with Drawing 67F2011 for single and Drawing 71236 for dual BAK-12 and shall consist of at least the following major components, arranged essentially as shown thereon. Additional components, including hardware, required to complete the aircraft arresting system for this particular type installation shall be as shown on Drawings 67F2011 and 71236.

Component	See Paragraph	Number Required
Energy absorber	3.8	2
Purchase element tapes	3.9	2
Purchase element connectors	3.10	2
Hook cable	3.11	1
Fairlead unit assemblies (special installations only)	3.12	2
Runway edge sheaves	3.13	2
Fairlead tubes (if required)	3.15	As specified
Tape brush assemblies	3.16	2
Tape connector adapter assembly (dual BAK-12)	3.17	2
Special service tools and kits	3.18	1 set.

* 3.5.3 Installation, permanent, pit type. The design of the aircraft arresting system shall be adaptable to installation in an underground concrete pit having a top which is substantially flush with the surrounding terrain. Arresting systems for pit-type installation shall be in accordance with Drawing 67F2012 for single and Drawing 68F211 for dual BAK-12 and shall consist of at least the following major components, arranged essentially as shown thereon. Additional components, including hardware, required to complete the arresting system for this particular type installation shall be as shown on Drawings 67F2012 and 68F211.

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Component	See Paragraph	Number Required
Energy absorber	3.8	2
Purchase element tapes	3.9	2
Purchase element connectors	3.10	2
Hook cable	3.11	1
Runway edge sheaves	3.13	2
Deflector sheaves	3.14	4
Fairlead tubes	3.15	As specified
Tape brush assemblies	3.16	2
Tape connector adapter assembly (dual BAK-12 only)	3.17	2
Special service tools and kits	3.18	1 set.

* 3.5.4 Reliability. The aircraft arresting system shall have a MCBF of 76 consecutive average arrestments, excluding replacement of purchase element tapes, absorber brake, and hook cable as needed. Reliability shall be a minimum of 97 percent at a confidence factor of 0.90 (see 6.3). Stipulation of average arrestments shall be interpreted to include any combination of aircraft weight and speed equal to approximately 40 x 10^6 foot-pounds for single 950-foot runout, 55 x 10^6 foot-pounds for 1,200-foot runout, and 100 x 10^6 foot-pounds for dual BAK-12.

3.5.5 Maintainability. The aircraft arresting system shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity.

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e. Minimum number of tools required for maintenance by design practice such as reducing the variety of bolthead sizes to the practicable minimum.

f. Daily servicing should require no more than 2-1/2 man-hours per day with an additional 2 hours for post arrestment servicing. Monthly inspection and functional checkouts should require no more than 9 man-hours. Replacing the hook cable should require no more than two men, 1 man-hour each. Replacing the purchase elements tapes should require no more than two men, 3 man-hours each.

3.6 Construction. The aircraft arresting system shall be so constructed that no parts will work loose in service. It shall be built to withstand the loads, strains, vibrations, and other conditions incident to service, shipping, and storage. (Reference to parts working loose in service includes premature functioning of parts, devices, or mechanisms which degrade system performance or cause equipment to operate improperly.)

3.7 Performance

3.7.1 Environmental conditions. The aircraft arresting system shall be capable of operating satisfactorily and without damage under the following environmental conditions:

a. Temperatures ranging from -40° to +125°F

b. Humidity up to 100 percent

c. Altitude, approximately 10,000 feet

d. Fungus

e. Rain

f. Salt-fog

g. Dust

h. Shock

i. Vibration.

3.7.1.1 The aircraft arresting system shall also be capable of withstanding, without damage, the environmental conditions specified in 3.7.1 for long periods of time in the standby status. In addition to the above, the arresting system shall be capable of withstanding an altitude of 50,000 feet in a transit status.

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* 3.7.2 Operational performance. The aircraft arresting system shall meet or exceed the following performance requirements with regard to energy absorption in relation to speed/weight for aircraft arrestments:

a. When the arresting system is installed to provide a system runout distance of 950 feet and having a 150- to 200-foot runway span, a split distance of 50 to 300 feet between runway edge sheaves and the energy absorber and using a 1-inch-diameter hook cable; the system shall:

(1) Have an energy capacity up to 65 x 10^6 foot-pounds

(2) Withstand hook engagements off-center up to 40 percent of semispan at the following weights and speeds:

Aircraft Weight (Pounds)	Engaging Speed (Knots)
20,000	190
25,000	190
30,000	190
35,000	190
40,000	190
45,000	186
50,000	176
55,000	167
60,000	158

(3) Withstand hook engagements off-center up to 80 percent of semispan with not more than a 15 percent reduction in the above speeds

(4) Not impose hook loads exceeding the following:

Aircraft Weight (Pounds)	Engaging Speed (Knots)	Maximum Hook Load (Pounds)
20,000	120	31,000
	140	38,500

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Aircraft Weight (Pounds)	Engaging Speed (Knots)	Maximum Hook Load (Pounds)
20,000	160	49,500
	180	65,000
30,000	120	35,000
	140	45,000
	160	57,000
	180	72,500
40,000	120	38,500
	140	49,500
	160	62,500
50,000	120	49,500
	140	62,500
	150	71,500
60,000	100	48,500
	120	60,500
	140	78,000

(5) Be capable of accepting 12 arrestments per hour

b. When the arresting system is installed to provide a system runout distance of 1,200 feet and having a 150- to 200-foot runway span, a split distance of 50 to 300 feet between runway edge sheaves and the energy absorber, and using a 1-1/4-inch-diameter hook cable; the system shall:

* (1) Have an energy capacity of 90 x 10^6 foot-pounds

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(2) Withstand hook engagements off-center up to 40 percent of semispan at the following weights and speeds:

Aircraft Weight (Pounds)	Engaging Speeds (Knots)
70,000	168
80,000	158
90,000	149
100,000	140
110,000	133

* c. When the arresting system is installed in the dual mode and provides a system runout distance of 1,200 feet and when equipped with a 1-1/4 inch diameter steel hook cable, the system shall:

(1) Have an energy capacity up to 170 x 10^6 foot-pounds

(2) Withstand hook engagements off-center up to 40 percent of semispan at the following weights and speeds:

Aircraft Weight (Pounds)	Engaging Speed (Knots)
60,000	190
70,000	190
80,000	190
90,000	190
100,000	190
110,000	180
120,000	170

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(3) Not impose hook loads exceeding the following:

Aircraft Weight (Pounds)	Engaging Speed (Knots)	Maximum Hook Load (Pounds)
60,000, 75,000, and 90,000	120	72,000
	140	85,000
	160	100,000
	180	115,000
	190	125,000
100,000	120	78,000
	140	92,000
	160	109,000
	180	129,000
	190	140,000
120,000	120	84,000
	140	102,000
	160	125,000
	170 (maximum allowable speed at this weight)	140,000

d. Under all conditions above, the tensile load in the purchase system shall not exceed 60 percent of the strength of the purchase system at its weakest points

e. The unbalanced steering force exerted upon an aircraft in off-center hook engagements shall be zero or shall have a moderate tendency to steer the aircraft toward center during runout.

3.8 Energy absorber. The energy absorber shall provide a maximum system runout of 1,200 feet and shall be capable of being programmed for a maximum system runout of 950 feet. The energy absorber shall be designed in accordance

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with Drawing 52-W-2291-2 and details thereto. A cam-controlled valve shall be incorporated to provide programmed brake pressure for a given aircraft weight. The control valve shall be provided with a variable adjustment to allow setting control for aircraft weights varying approximately between 20,000 and 60,000 pounds for a 950-foot system runout, and shall be adjustable for
* aircraft weights between 70,000 and 120,000 pounds for a 1,200-foot system runout. The control system shall automatically accommodate increases or decreases in engaging speed by a corresponding increase or decrease in retarding force.

3.8.1 Hydraulic control system. The hydraulic control system for the energy absorber shall operate at pressures up to 3,000 psi at suitable factors of safety for all hydraulic components.

3.8.1.1 Hydraulic pressure. A pressure relief valve shall be incorporated to limit the hydraulic control pressure within the safety factors. A pressure gage conforming to MIL-G-8402 shall be provided on each absorber to indicate the static pressure and the maximum brake pressure reached during an engagement.

3.8.2 Rewind system. The energy absorber shall be provided with a system to rewind the purchase element tapes on the storage reel after an arrestment and also provide a pretension on the purchase system after rewind of 1,500 to 2,000 pounds tension. It shall consist essentially of a gasoline engine complete with accessories and controls, fluid coupling, gear reducer, and a rewind clutch. The rewind engine also shall be used to provide a precise rotation to the main pump of the hydraulic braking system during the process of synchronizing the control system (see 6.3.1). The gasoline engine shall conform to the group 2 requirements of MIL-E-11275. The engine shall have sufficient horsepower (approximately 35 hp) and torque (90 foot-pounds at 1,400 rpm) to return the purchase system after a 950-foot runout to the battery position in not more than 3 minutes (1,200 feet in 3.78 minutes) and to meet pretension requirement cited above.

3.8.3 Absorber base. The absorber shall be modified as outlined on Drawing 52-C-963-1 for concrete pad or pit installation.

3.9 Purchase element tapes. Purchase element tapes (purchase tapes) shall be in accordance with MIL-W-38461, type I. The purchase element length will be variable due to system runout, runway span, and the distance between the runway edge sheaves and energy absorber. Purchase element tape length shall be as specified for each installation (see 6.2).

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3.10 Purchase element connectors. Purchase element connectors for connecting the purchase element tapes to the hook cable shall be as follows:

Hook Cable Diameter	Drawing No.
1 inch	52-W-2081-1
1-1/4 inches	66D1751.

3.11 Hook cable. The hook cable (arresting pendant) shall be as follows:

System Runout	Cable Diameter	Drawing No.
950 feet *	1 inch	62B35556
1,200 feet	1-1/4 inches	515053

* Special installations only, see 3.5.e.

The hook cable length will be variable depending on the runway span and shall be as specified (see 6.2).

3.12 Fairlead unit assemblies. Fairlead unit assemblies (Fairlead Beams) shall be in accordance with Drawing 52-W-2295-1 for soil-surface installation. (May be used for concrete pad special installations. When so used, it shall be modified in accordance with Drawing 67F2011.)

* 3.13 Runway edge sheaves. Runway edge sheaves (deck sheaves) for surface and pit type permanent installation single BAK-12 shall be in accordance with Drawing 52-W-2080-1. Runway edge sheaves for surface and pit type permanent installation dual BAK-12 shall be in accordance with Drawings 52-W-2080-1 and 52-W-2252-1.

3.14 Deflector sheave assemblies. Deflector sheave assemblies used in concrete pit installations shall be in accordance with Drawing 52-D-380-1. Installation and mounting shall be as shown on Drawing 67F2012.

3.15 Fairlead tubes and couplings. Fairlead tubes and couplings used between runway edges and the energy absorber to protect purchase elements from the effect of weather normally will be furnished by the Government. When specified (see 6.2), the Fairlead tubes and couplings shall be furnished by the contractor. Fairlead tubes and couplings shall be as shown on Drawing 67F2011 or 67F2012.

3.16 Tape brush assemblies. Separate tape brush assemblies for cleaning the purchase element tapes are required when runway edge sheaves are used. Tape brush assemblies are built into the Fairlead unit assemblies. The tape brush assemblies shall be in accordance with Drawing 52-D-109-1.

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The tape brush mounting for 10-3/4-inch OD Fairlead tubes shall be in accordance with Drawing 52-D-642-1 as modified by Drawings 67F2011 and 67F2012. If other size Fairlead tubes are used, this must be further modified.

3.17 Tape connector adapter assemblies (for dual BAK-12 only). The tape connector adapter assemblies are required to convert a single BAK-12 into a dual BAK-12 arresting system. A complete assembly consists of an adapter (71224-01), pin (71225-01), nut (71226-05), and a setscrew (see Drawing 68F211 or 71236 for detail description of the setscrew). Two adapters are required for a dual BAK-12 arresting system.

3.18 Special service tools and kits. Unless otherwise specified (see 6.2), special service tools and kits for installation and maintenance of the arresting system shall be supplied with the system. The special service tools and kits shall contain at least the following:

a. Purchase element tape punch

b. Synchronizing chain in accordance with Drawing 52-W-2288-16

c. Stake driving tool (for soil surface installations only) in accordance with Drawing 52-B-1244

d. Bearing puller tool in accordance with Drawing 52-B-624-2

e. Hook cable support disc installation kit consisting of loading arbor, Drawing 64C33478-11, and loading tube, Drawing 64C33518-11, for use with the 1-1/4-inch cable. When the system is designated for special installation per 3.5e, loading arbor, Drawing 64C33478-7, and loading tube, Drawing 64C33518-7, is required for 1-inch cable

f. Accumulator charging kit in accordance with Drawing 52-C-1246-1

g. Brake bleed kit in accordance with Drawing 52-C-1247-1

h. Synchronizing kit in accordance with Drawing 52-C-1248-1

i. Manual hydraulic pump with base assembly in accordance with Drawings 152-A-1092 and 152-A-1093.

3.19 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.20 Electromagnetic interference characteristics. Electromagnetic interference control requirements shall be in accordance with MIL-STD-826, method 7001.

3.21 Finishes and protective coatings. Cleaning, painting, plating, chemical films, and anodic coatings shall be in accordance with MIL-STD-808. All painted surfaces shall be painted for type I exposure. The colors shall be as follows:

a. Exterior surfaces - Color code DG (yellow)

b. Interior surfaces - Color code LG (aircraft gray)

c. Equipment within enclosure - Color code LG (aircraft gray).

3.22 Markings

3.22.1 Nameplates, instruction plates, data plates, and placards. All nameplates, instruction plates, data plates, and placards shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required. Abbreviations shall be in accordance with MIL-STD-12. Mechanical symbols shall be in accordance with MIL-STD-17. All such plates shall be attached by metal screws or rivets.

3.22.1.1 Vendor's nameplate. Only one vendor's nameplate, not exceeding 6 by 9 inches in size, shall be installed on the arresting system.

3.22.2 Operational markings. Operational markings and stenciling shall be in accordance with the requirements of MIL-STD-808.

3.22.3 Identification of controls and instruments. Where applicable, all gages, controls, and instruments used in the operation of the arresting system shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags shall not be used.

3.22.4 Operating instructions. Where applicable, brief operating and precautionary instructions shall be permanently affixed on or near the control panel or other points as necessary. The instructions shall be clear, concise, and adequate to enable operation without damage to the system or injury to operating personnel. Precautionary instructions shall be so designed and located that the operator cannot fail to notice them before controls, valves, and switches are operated.

3.23 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.24 Workmanship. The aircraft arresting system, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of all parts, and tightness of assembly screws and bolts.

3.24.1 Cleaning. The aircraft arresting system shall be thoroughly cleaned and loose, spattered, or excess solder; metal chips; and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the aircraft arresting system shall be classified as follows:

a. Preproduction testing . . . . . See 4.3

b. Acceptance tests . . . . . See 4.4.

4.3 Preproduction testing (see 6.2)

4.3.1 Test sample. One aircraft arresting system shall be subjected to the tests specified in 4.3.3.

4.3.2 Test report. After completion of preproduction testing, a test report shall be prepared in accordance with MIL-STD-831 and three copies furnished to the procuring activity.

4.3.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, and miles or position in the test procedure, as appropriate. Test conditions at

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b. A summary of the engineering analysis and tests conducted to determine assignable causes for any failure or irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control or other procedures, or leading to any corrections or changes proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such operations.

d. Clock time and man-hours required for each maintenance and servicing action during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included and shall be adequate to permit comparison to the personnel anticipated in similar field work.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.3.3 Preproduction tests. Preproduction tests shall consist of all the tests described under 4.5.

4.3.3.1 Preproduction testing by the contractor. Unless otherwise specified, the contractor shall subject one aircraft arresting system to all tests under 4.5, except 4.5.4 and 4.5.6. The contractor shall then submit the arresting system to the procuring activity for the tests specified in 4.5.4 and 4.5.6.

4.4 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.4.1 Individual tests. Each aircraft arresting system shall be subjected to the following tests as described under 4.5:

a. Examination of product . . . . . See 4.5.1

b. Mechanical inspection . . . . . See 4.5.2.

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4.5 Test methods

4.5.1 Examination of product. The aircraft arresting system shall be inspected to determine compliance with the requirements of this specification. All features and components shall be inspected with respect to materials, workmanship, dimensions, proper installation, and marking.

4.5.2 Mechanical inspection. A critical mechanical inspection shall be made of all components and parts of the aircraft arresting system to insure proper functioning of all mechanical parts, accessibility of controls and adjustments, installation, and to demonstrate interchangeability of removable parts without special matching. All components shall be moved through their maximum normal displacement and shall exhibit no malfunctioning, leakage, objectionable rough operation, or irregular movements. The inspection shall include complete lubrication and servicing of the energy absorber unit to insure readiness for operation.

4.5.3 Environmental tests. The aircraft arresting system shall be subjected to the following tests conducted in accordance with the specified methods of MIL-STD-810:

a. High temperature - method 501, procedure I, except the temperature shall be 125°F

b. Low temperature - method 502, procedure I, except the temperature shall be -40°F

c. Humidity - method 507, procedure I, except the temperature shall be 125°F

d. Fungus - method 508, procedure I

e. Salt-fog - method 509, procedure I

f. Rain - method 506, procedure I

g. Dust - method 510, procedure I, except the temperature shall be 125°F.

4.5.4 Operational test (to be conducted by the procuring activity). The aircraft arresting system shall be functionally tested against manned aircraft to determine compliance with the requirements specified herein. These tests shall be conducted over a full range of aircraft weights and speeds as specified in 3.7.2 and shall include engagements made on-center and off-center up to 80 percent of runway semispan.

4.5.4.1 Energy absorber temperature. The arresting system shall be engaged at short cycle intervals for 12 high-energy arrestments in 1 hour, during which period no critical energy absorber component shall exceed 275°F.

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4.5.5 Electromagnetic interference. The aircraft arresting system shall be tested for electromagnetic interference in accordance with MIL-STD-826, method 7001.

4.5.6 Reliability demonstration and testing (to be conducted by the procuring activity). The entire arresting system shall be installed and tested under operational conditions at a test site or operational base designated by the contracting officer. This testing shall be accomplished by the procuring activity. However, the contractor shall provide on-site technical support personnel as specified in the contract (see 6.2).

4.5.7 Maintainability evaluation. The aircraft arresting system shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.6 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Disassembly. Disassembly of the system components shall be accomplished to the extent practicable to reduce cube and facilitate packaging and packing of the system.

5.1.1.2 Cleaning and drying. Component parts of the aircraft arresting system shall be cleaned in accordance with MIL-P-116, process C-1, and dried in accordance with MIL-P-116, procedure D-1 or D-4.

5.1.1.3 Preservation application. Immediately after cleaning and drying, all exposed bare metal surfaces and critical parts (ie, sprockets, gears, chain drives, et cetera) shall be coated with P-1 or P-2 preservative in accordance with MIL-P-116, method I. Where ease of removal is not a prime consideration, P-1 preservative should be used.

5.1.1.4 Engines. Engines shall be preserved and packaged in accordance with MIL-E-10062.

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5.1.1.5 Electrical components. Electrical components shall be preserved in accordance with MIL-E-17555.

5.1.1.6 Instrument panel. Glass on instrument panel shall be covered with PPP-F-320 or other similar protective material and secured with tape conforming to PPP-T-60.

5.1.1.7 Hook cable. The hook cable (pendant) shall be preserved and packaged in accordance with MIL-STD-163.

5.1.1.8 Purchase element tapes. Purchase element tapes shall be packaged in accordance with requirements of MIL-W-38461 for type 1 webbing, except as modified in 5.2.1.2 herein.

5.1.1.9 Tools. Special service tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.1.10 Kits. All kits supplied with the system shall be preserved and packaged in accordance with MIL-STD-794, appendix E.

5.1.2 Level C. Components of the aircraft arresting system shall be preserved and packaged in a manner which affords adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Components of the system, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1.1 through 5.2.1.10.

5.2.1.1 Energy absorber. Each energy absorber shall be packed in a container conforming to MIL-C-104, type I, class 2, style A.

5.2.1.2 Purchase element tapes. Purchase element tapes shall be packed in accordance with requirements of MIL-W-38461 for type 1 webbing, except reels shall be modified to provide for skids and rubbing strips to provide ease of fork lift entry.

5.2.1.3 Fairlead unit assemblies. Fairlead unit assemblies shall be packed in containers conforming to MIL-C-3774.

5.2.1.4 Hook cable. Hook cables (pendants) shall be packed in accordance with MIL-STD-163.

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5.2.1.5 Purchase element connectors. Purchase element connectors shall be packed in containers conforming to PPP-B-601, overseas type, style A or B.

5.2.1.6 Runway edge sheaves and deflector sheave assemblies. Runway edge sheaves and deflector sheave assemblies shall be packed in containers conforming to MIL-B-26195, type II.

5.2.1.7 Fairlead tubes and couplings. Fairlead tubes and couplings, when specified (see 6.2), shall be packed in containers conforming to PPP-B-621, class 2.

5.2.1.8 Tape brush assemblies. Tape brush assemblies shall be packed in containers conforming to PPP-B-601, overseas type, style A or B.

5.2.1.9 Special service tools. Tools shall be packed in accordance with PPP-P-40.

5.2.1.10 Kits. Kits supplied with the system shall be packed in accordance with MIL-STD-794, appendix E.

5.2.1.11 Consolidation, limitations, and closure. Components shall be consolidated into the fewest number of crates practicable consistent with required protection. The gross weight of each pack shall not exceed weight and size limitations specified in the applicable container specification. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto.

5.2.2 Level B. Level B packing shall be the same as level A except domestic type containers, where available, in accordance with the applicable container specifications shall be used in lieu of overseas type containers.

5.2.3 Level C. Components of the aircraft arresting system, preserved and packaged as specified in 5.1.2, shall be packed in a manner which affords adequate protection against damage during shipment from the supply source to the first receiving activity. Shipping containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation. This may conform to the supplier's commercial practice when it meets the requirements of this level.

5.3 Physical protection. Interior cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186.

5.4 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. In addition, to facilitate reassembly, mating parts that are removed from the basic assembly shall be matchmarked

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with lusterless white enamel and overcoated with water-resistant clear spar varnish conforming to TT-V-121. Small parts that cannot be matchmarked as specified above shall be matchmarked with waterproof tags conforming to UU-T-81, type A or B, class 2.

6. NOTES

6.1 Intended use. The BAK-12A/E32A aircraft arresting system is intended for emergency arrestment of hook equipped aircraft at speeds up to 190 knots at energy levels up to 65 million foot-pounds for a 950-foot runout and up to 85 million foot-pounds for a 1,200-foot runout.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. In the event it becomes necessary to change the drawings (see 3.5) in order to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract

c. Type of installation for which each arresting system is to be used (see 3.5, 3.5.1, 3.5.2, 3.5.3) including:

(1) Length of hook cable (see 3.11)

(2) Span between runway edge sheaves

(3) Split distance between runway edge sheaves and the energy absorber

(4) If only a 950-foot system runout is required and if a 1-inch-diameter hook cable will be allowed

* (5) When dual absorbers are to be used

d. For soil surface installations (see 3.5.1)

(1) When reinstallation kit is to be supplied (see 3.5.1.1)

(2) When special expeditionary installation kits are to be supplied and contents of these kits (see 3.5.1.2)

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e. For permanent, surface type installations (see 3.5.2)

(1) When special installation will use Fairlead unit assemblies in place of runway edge sheaves (see 3.12)

(2) When Fairlead tubes and couplings are to be supplied by the contractor (see 3.15)

f. For permanent, pit type installations (see 3.5.3)

(1) When Fairlead tubes and couplings are to be supplied by the contractor (see 3.15)

(2) Options as specified on the drawings

g. Purchase element tape length for each arresting system (see 3.9)

h. When special service tools and kits are not to be supplied with the arresting system (see 3.18)

i. Location and conditions for preproduction testing including

(1) Tests to be conducted by the contractor (see 4.3.3.1)

(2) Tests to be conducted by the procuring activity (see 4.3.3.1)

(3) Extent of contractor support and participation during preproduction testing by the procuring activity (see 4.5.4 and 4.5.6)

j. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Definitions

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified number of cycles.

6.3.2 Mean cycles between failures (MCBF). The mean life or MCBF is the arithmetical mean (average) of the operating cycles between failures.

6.3.3 Runway span. Runway span is the cross-runway distance between the systems runway edge sheaves. This distance does not necessarily coincide with the runway width.

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6.3.4 Split distance. Split distance is the distance between the runway edge sheave and the energy absorber on a given side. This distance is measured from the geometric center of the runway edge sheave to the edge of the energy absorber nearest the runway edge sheave.

6.3.5 Synchronization brake pressure. To insure that the energy absorbers are hydraulically balanced and that each energy absorber has the correct weight setting, it is necessary to synchronize both energy absorbers after installation or replacement of components that could affect braking of reel. This operation entails the use of the rewind engine to drive the main hydraulic pump providing brake pressure to reels. Using the cam linkage mechanism, a particular brake pressure is obtained at a reference location on the cam, while the main pump is rotating at a given rpm. (See 3.8.2.)

* 6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 1710-F017

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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